

Retrofitting aging gas or oil boilers in small commercial properties with a scalable, highly efficient R290 heat pump system providing space heating and domestic hot water.

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£4,500.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

10

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source
- [D] Algorithmic estimate or LLM extraction
- [E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	PASS	Standards found	5 standards
3. Cost Ceiling Specified	PASS	Target provided	£4,500.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	10 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements [A]

1.1 Mission & Use Case

Mission

To accelerate the decarbonization of UK SME estates by providing an affordable, high-efficiency, R290-based 30kW commercial heat pump.

Use Case

Retrofitting aging gas or oil boilers in small commercial properties with a scalable, highly efficient R290 heat pump system providing space heating and domestic hot water.

1.2 Market Context & Timing [B]

The UK's Net Zero 2050 mandate, shifting F-Gas regulations drastically limiting traditional refrigerants by 2030, and soaring fossil fuel costs make ultra-efficient, low-GWP commercial heating a strictly timed necessity.

Small commercial landlords, boutique hotels, medium-sized retail spaces, office building facility managers, and commercial HVAC installers.

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Vaillant

Product	aroTHERM plus (Cascade)
Pricing	Approx £12,000 for a 3-unit cascade system to reach 30kW.
Tech Specs	R290 refrigerant, max 12kW single unit (requires cascading for 30kW), COP ~ 3.6 to 4.0 at A2/W35, Monobloc design.

STRENGTHS

Highly trusted UK brand, excellent reliability, ultra-low GWP, very quiet operation.

WEAKNESSES

Requires cascading multiple smaller units to reach 30kW which increases installation footprint, piping complexity, and total cost.

DIFFERENTIATION ANGLE

Providing a single, compact 30kW unit significantly reduces absolute footprint, hydraulic complexity, and installation time compared to cascaded domestic units.

Mitsubishi Electric

Product	Ecodan CAHV
Pricing	Approx £15,000 unit cost.
Tech Specs	43kW capacity, R407C (high GWP), cascade capable, COP ~ 3.4 at A2/W35.

STRENGTHS

Dominant market presence, extremely robust commercial warranty support, proven long-term durability in UK climate.

WEAKNESSES

Uses phased-out/high-GWP refrigerants, high unit cost, larger capacity than needed for small 30kW footprint.

DIFFERENTIATION ANGLE

Use of R290 future-proofs the product against F-Gas bans, matches the specific 30kW sweet spot for small premises, and hits a far more aggressive £4,500 capex price point.

Clivet

Product	Thunder (Commercial R290)
Pricing	Approx £9,500 - £12,000 unit cost.
Tech Specs	R290 refrigerant, 30kW - 80kW range, COP ~ 3.3 at A2/W35, Packaged outdoor monobloc.

STRENGTHS

Dedicated commercial design, compliant with latest European ecological standards, good BMS integration.

WEAKNESSES

High upfront capital cost restricts adoption by smaller SMEs, lower COP at required testing conditions.

DIFFERENTIATION ANGLE

Targeting a highly aggressive BOM/Unit cost of £4,500 through standardized tier-2 commercial components to monopolize the cost-sensitive UK commercial retrofit market.

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£4,500.00
Max Mass	Not computed

Research Sources

Title	Type	Year	Relevance
BSRIA UK Heat Pump Market Report 2023-2024	Market Research	2026	Provides quantitative market size, unit volume expectations, and price-point benchmarking for commercial heat pumps in the UK.
MCS 020 Standard for Heat Pumps	Regulatory Standard	2026	Outlines the precise acoustical and performance requirements necessary for commercial grant eligibility in the UK.
CIBSE Guide B1: Heating	Engineering Guide-line	2026	Key reference for commercial hydronic system design, flow temperatures, and thermal load matching in UK buildings.
F-Gas Phase Down Schedule (UK/EU)	Legislation	2026	Dictates the timeline for the phase-out of HFCs, validating the strategic shift to R290 (Propane).
Danfoss & Copeland R290 Compressor Catalogs	Supplier Architecture Data	2026	Cost and technical specification sourcing required to validate the £4,500 unit cost at 500 units/year volume.

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2.1 MCS 020 [D]

Microgeneration Certification Scheme - Planning

Status: not-started

Owner: Compliance Engineer

Summary

Establishes sound level limits and performance standards for heat pumps to be permitted installations and grant-eligible.

Applicability

Applies to the entire outdoor unit assembly and performance claims.

Engineering Impact

Requires rigorous sound testing and potentially acoustic baffling on the fan/compressor; dictates verified COP claims.

EVIDENCE REQUIRED

Independent acoustic and performance testing reports from a UKAS accredited lab.

GAP ACTION

Schedule acoustic pre-compliance testing with an external laboratory.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.2 EN 378 [D]

Safety and environmental requirements for refrigerating systems

Status: not-started

Owner: Lead Mechanical Engineer

Summary

Covers safety requirements for the design, construction, installation, and operation of refrigerating systems, specifically addressing flammability.

Applicability

Applies to the R290 refrigerant circuit, charge limits, and outdoor/indoor module safety separation.

Engineering Impact

Mandates A3 (highly flammable) safety measures, including spark-free components, explosion-proof enclosures for electricals, and leak detection protocols.

EVIDENCE REQUIRED

FMEA documentation, ATEX-rated component certificates, and system safety design dossier.

GAP ACTION

Conduct hazardous area classification review and component specification audit.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.3 BS EN 14511 [D]

Air conditioners, liquid chilling packages and heat pumps criteria

Status: not-started

Owner: Thermal Systems Engineer

Summary

The mandated testing standard for determining the heating capacity and COP (Coefficient of Performance).

Applicability

Overall thermodynamic performance (Target: 30kW, COP >= 3.5 @ A2/W35).

Engineering Impact

Drives the sizing of the evaporator, condenser, and compressor selection to guarantee rated outputs.

EVIDENCE REQUIRED

Performance test results at A2/W35 under standardized laboratory conditions.

GAP ACTION

Build a functional thermodynamic prototype for initial internal calorimeter screening.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.4 PED 2014/68/EU [D]

Pressure Equipment Directive

Status: not-started

Owner: Manufacturing Engineer

Summary

Regulates the design, manufacture, and conformity of pressure equipment operating over 0.5 bar.

Applicability

Piping, heat exchangers, compressor, and refrigerant receivers operating under high pressure.

Engineering Impact

Requires burst-pressure calculations, specific brazing qualifications, and pressure relief valve integration.

EVIDENCE REQUIRED

Material certificates, pressure test reports, and
Notified Body design examination.

GAP ACTION

Compile Bill of Materials and verify pressure
ratings and supplier PED certificates.

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.5 Part L [D]

UK Building Regulations: Conservation of Fuel and Power

Status: not-started

Owner: Product Manager

Summary

Mandates energy efficiency standards for new and existing commercial non-domestic buildings in the UK.

Applicability

Targeted at the final installed system; influences customer buying decisions.

Engineering Impact

Requires the unit to easily interface with modern building energy management systems (BEMS) and weather compensation controls.

EVIDENCE REQUIRED
Product ErP (Energy-related Products) label and seasonal space heating energy efficiency (SCOP) data.

GAP ACTION
Develop the control board interface specifications for BEMS/Modbus connectivity.

Data Sources
[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

3. Sizing & Spatial Optimisation [D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m²
Internal Volume	75 m³
Rules Domain	hvac_and_renewables

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m²)	Mount
outdoor_structural_chassis	2236 × 2236 × 2400	5	floor
air_heat_recovery_module	2236 × 2236 × 2400	5	floor
r290_compressor_assembly	2236 × 2236 × 2400	5	floor
refrigerant_valving_circuit	2236 × 2236 × 2400	5	floor
thermal_transfer_condenser	2236 × 2236 × 2400	5	floor
outdoor_power_control_panel	2236 × 2236 × 2400	5	floor
indoor_hydrobox_chassis	2236 × 2236 × 2400	5	floor
hydronic_circulation_assembly	2236 × 2236 × 2400	5	floor
backup_heating_element	2236 × 2236 × 2400	5	floor
system_controller_interface	2236 × 2236 × 2400	5	floor

Conflicts

- Total module footprint (50.0m²) exceeds envelope budget (21.3m²)

Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 Outdoor Structural & Acoustic Chassis [D]

Maturity: draft

Purpose

Provides the weather-resistant physical framework for the outdoor R290 unit and attenuates compressor/fan noise to meet MCS regulations.

Why It Matters

Without robust weatherproofing and acoustic attenuation, the unit will fail UK MCS 020 sound certifications, locking the product out of highly lucrative commercial boiler upgrade grants.

Technical Description

This structural framework forms the physical backbone of the outdoor unit, designed to withstand harsh UK weather conditions for 15+ years while safely segregating the highly flammable R290 components. The physical layout must ensure that in the event of a refrigerant leak, heavier-than-air propane gas safely diffuses to the environment without pooling near electrical ignition sources, meeting strict ATEX / DSEAR explosive atmosphere directives.

Beyond basic structural integrity and crash-resistance for shipping, this module acts as the primary acoustic barrier. To meet MCS 020 sound level requirements for commercial grant eligibility, the chassis integrates strategically placed acoustic baffles, low-frequency resonance dampeners, and heavy EPDM rubber isolation mounts. This tuning mitigates the low-frequency humming produced by the 30kW compressor and the aerodynamic noise of the large EC fans, making the unit suitable for dense commercial and urban installations.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 Air Heat Recovery Assembly (Evaporator) [D]

Maturity: draft

Purpose

Extracts low-grade thermal energy from the ambient air and transfers it to the R290 refrigerant.

Why It Matters

The efficiency of this module dictates the system's overall Coefficient of Performance (COP). Inadequate thermal extraction here directly causes the system to miss the critical e 3.5 COP target, rendering it uncompetitive.

Technical Description

This sub-assembly acts as the thermal engine of the heat pump, physically pulling vast quantities of ambient air across a dense network of copper tubes and aluminum fins to boil the R290 refrigerant. Because maintaining a COP of e 3.5 at A2/W35 is the central performance metric, the surface area and geometry of the finned coil are maximized to minimize the approach temperature between the air and the refrigerant. The hydrophilic-coated aluminum fins rapidly shed condensation, optimizing airflow and reducing static pressure drop.

In the damp, cool UK climate, outdoor evaporators are highly susceptible to frost accumulation, which severely drastically degrades thermal transfer. Consequently, this module incorporates precision-controlled Electronically Commutated (EC) axial fans that adjust RPM dynamically based on temperature differentials. It also integrates a heavily sloped defrost drain pan lined with self-regulating trace heating tape to ensure that defrosted ice (meltwater) completely voids the system, preventing fatal ice-expansion damage to the lower copper coils during deep-winter operation.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 R290 Compressor Subsystem [D]

Maturity: draft

Purpose

Compresses low-pressure R290 vapor into high-pressure, high-temperature vapor, driving the thermodynamic cycle.

Why It Matters

This is the most expensive and critical rotating component of the heat pump. Reliability, turndown ratio, and efficiency here make or break the unit's 15-year commercial viability and CAPEX targets.

Technical Description

The compressor subsystem takes the low-temperature vapor boiled by the evaporator and significantly increases its pressure, fundamentally elevating its thermal energy to a temperature high enough to provide 70°C+ output water. Strategic selection of a Tier 1 (e.g., Copeland) inverter-driven scroll compressor optimized specifically for R290 is critical, as propane has distinct compression ratios and discharge temperatures compared to legacy refrigerants like R32 or R410A. The variable frequency drive (VFD) integration ensures that the compressor seamlessly modulates output precisely to building thermal demands, preventing short-cycling.

Because R290 is extremely flammable, the compressor must be specially rated for A3 refrigerants, meaning internal electrical terminals are hermetically sealed and guaranteed spark-free. Additionally, compressors are the primary source of operational acoustic noise. This module must package the compressor in a tailor-made, heavy-mass acoustic jacket and implement suction accumulators and discharge mufflers to smooth out pressure pulsations. A crankcase heater is also integrated to prevent the highly soluble propane liquid from pooling in the compressor polyolester (POE) oil during cold stand-by periods.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Refrigerant Routing & Control Circuit [D]

Maturity: draft

Purpose

Dynamically regulates the pressure and directional flow of the extremely flammable R290 refrigerant across operating conditions.

Why It Matters

Strict PED compliance and perfectly brazed joints are a legal mandate. Any catastrophic leak here not only ruins performance but introduces a massive explosion hazard into a commercial environment.

Technical Description

This critically sealed pipework and valving module orchestrates the physical fluid dynamics of the highly flammable R290 circuit. The central component is the Electronic Expansion Valve (EEV), a highly precise, stepper-motor-driven orifice that throttles the high-pressure liquid propane down to a low-pressure, super-cooled mist just before it enters the evaporator. The module also contains the 4-way reversing valve, an essential component for the UK environment that temporarily reverses the cycle to flood the outdoor evaporator with hot gas to melt winter ice accumulation.

Due to the volatile nature of the A3 refrigerant, this entire sub-assembly demands rigorous adherence to the Pressure Equipment Directive (PED 2014/68/EU). All copper piping must be sized flawlessly, silver-brazed by certified technicians, and burst-tested mathematically and physically. Precision high and low-pressure transducers continually monitor fluid state and report to the central controller, ensuring operation remains safely inside the compressor's thermodynamic envelope. The entire pipework is heavily insulated to prevent efficiency drops and localized condensation.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 R290-to-Water Condenser Module [D]

Maturity: draft

Purpose

Safely transfers high-grade thermodynamic heat from the explosive outdoor R290 circuit into the safe, indoor-bound water/glycol loop.

Why It Matters

This module enables the entire product safety strategy. By restricting R290 to this outdoor heat exchanger, the unit completely avoids indoor explosive zone designations, maximizing market appeal and slashing installation costs.

Technical Description

Crucial to the 'hydronic split' architecture ensuring R290 never enters the occupied building, this module serves as the energetic bridge between the hazardous outdoor unit and the safe indoor premises. High-temperature propane gas (pulled from the compressor module) condenses on one side of a heavily reinforced, completely sealed stainless steel Brazed Plate Heat Exchanger (BPHE). On the other side of the corrugated plates, a continuous loop of water/glycol mixture flows, extracting the latent heat of condensation and sweeping it indoors at temperatures highly suitable for commercial radiators or underfloor systems.

Designing this interface requires precise thermodynamic calculation to optimize heat transfer while minimizing pressure drop on the hydronic side. Because it sits outdoors, freeze protection is a paramount engineering priority. The module integrates deep thermal jacketing, inline flow switches to prove circulation before compressor firing, and backup anti-freeze immersion thermostats that can trigger emergency circulation to prevent the BPHE waterways from freezing, cracking, and catastrophically mixing water into the sealed R290 loop.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 Outdoor ATEX-Isolated Master Controller [D]

Maturity: draft

Purpose

Manages unit algorithms, provides high-power VFD control to the compressor, and ensures electrical safety in an explosive atmosphere.

Why It Matters

This module ensures that any R290 leak does not result in a catastrophic commercial disaster. It guarantees regulatory EN 378 / ATEX compliance while driving the smart inverter logic necessary for high-COP operation.

Technical Description

This critically important module acts as the brain and power-distributor for the outdoor unit. It ingests 400V 3-phase commercial grid power and safely conditions it to drive the high-draw compressor via an advanced Variable Frequency Drive (VFD), while simultaneously managing all subsystem control logic (defrost algorithms, EEV tracking, safety limit validation). To achieve ultra-high efficiency, the master PCB constantly monitors arrays of thermistors and transducers, mapping thermodynamic states against optimized control curves stored in its firmware.

Due to the presence of 30kW of electrical switching hardware housed alongside a highly flammable A3 hydrocarbon, the electrical engineering here revolves strictly around BS EN 378 compliance. The entire electrical sub-assembly is segregated within an ATEX-rated, forced-ventilation enclosure. In the event an onboard ATEX R290 gas detector identifies a leak exceeding 15% of the Lower Flammability Limit (LFL), this panel is engineered to instantaneously physically isolate all non-intrinsically-safe circuits and trigger purging fans, effectively preventing spark-ignition.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 Indoor Hydrobox Enclosure & Manifold ^[D]

Maturity: draft

Purpose

Provides the safe, aesthetic indoor interface for the hydronic fluid logic, shielding operators and connecting to building infrastructure.

Why It Matters

This module represents the primary physical touchpoint for HVAC installers. If it is complex, cramped, or un-aesthetic, commercial adoption will stall. It also acts as the primary thermal insulator for the indoor distribution loops.

Technical Description

Since the volatile R290 refrigerant is confined outdoors, this indoor module functions as a safe, highly modular hydronic control center. Designed aesthetically to mimic a modern boiler, the fabricated steel chassis mounts to the wall of the commercial plant room or hallway. It acts as the structural carrier for all the interior hydronic circulation components, the auxiliary heaters, and the building management interfaces.

Great care must be taken regarding thermal insulation. The interior utilizes heavily molded Expanded Polypropylene (EPP) blocks that perfectly nest the internal hydronic module components. This prevents any ambient heat loss into the utility room when operating at high temperatures, and more importantly, stops exterior cabinet condensation during off-cycles or if the unit is optionally run in a chilled-water cooling mode during summer. The manifold arrangement provides quick-connect unions for standard UK commercial plumbing (e.g. 28mm or 35mm copper).

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 Primary Hydronic Pump & Safety Rack [D]

Maturity: draft

Purpose

Circulates the thermal transfer fluid between the outdoor condenser and the indoor commercial heating distribution system safely and efficiently.

Why It Matters

Without robust and perfectly modulated flow, the outdoor heat exchanger will overheat and fault, or the resulting COP will collapse due to parasitic pump power draw. It protects against highly damaging liquid overpressure.

Technical Description

This sub-assembly is the cardiovascular system of the heat pump architecture. Once the outdoor condenser heats the water/glycol mixture, the high-efficiency Variable Speed EC primary circulation pump physically drives that thermal mass deep into the commercial property. The pump speed dynamically matches the required delta-T (temperature difference) demanded by the system controller, ramping down when the building is up to temperature to aggressively save parasitic electrical energy.

Crucially, this module is packed with hydronic safety and maintenance components necessary to meet UK building regulations and to protect the expensive outdoor heat exchanger. A diaphragm expansion vessel absorbs the volumetric expansion of water as it hits 70°C. An automatic air deaerator actively removes micro-bubbles from the glycol, preventing pump cavitation and ensuring maximum thermal contact. Finally, a brass Pressure Relief Valve (PRV) safely vents pressure to a physical drain if system pressure exceeds design parameters, protecting against catastrophic pipe bursts inside the SME premises.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.9 Auxiliary Electric Boiler Backup ^[D]

Maturity: draft

Purpose

Provides emergency failsafe heating, thermal boosting during deep freeze days, and extremely high temperatures for legionella pasteurization.

Why It Matters

Ensures absolute heating reliability for risk-averse commercial clients, while also providing a cheap and highly reliable mechanism for mandatory DHW anti-legionella cycles.

Technical Description

While a 30kW R290 heat pump is highly capable in cold weather, commercial estates cannot afford any downtime if the outdoor unit requires maintenance, or if a freak cold snap (-15°C) exceeds the compressor's thermal capacity. This module is an inline resistive electrical heater built directly into the indoor hydrobox. It leverages solid-state relays (SSRs) to seamlessly pulse electrical elements, injecting exact increments of additional heat into the water stream without overwhelming the local circuit breakers.

Beyond backup scenarios, this module serves an essential health and safety role. UK commercial venues with domestic hot water (DHW) storage (like boutique hotels or clinics) are legally required to routinely pasteurize their water tanks above 60°C to kill Legionella bacteria. While R290 can hit these temperatures, utilizing the auxiliary heater to precisely and rapidly top-up the final few degrees during the pasteurization cycle takes significant mechanical strain off the expensive compressor and ensures compliance with strict UK water safety codes.

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.10 BEMS Gateway & Human Machine Interface (HMI) ^[D]

Maturity: draft

Purpose

Provides user control, installer diagnostics, and seamless integration with existing Building Energy Management Systems (BEMS) aligning with Part L regulations.

Why It Matters

Commercial facility managers require standard BMS integration; without Modbus/BACnet, the unit is un-sellable to mid-sized estates. An intuitive UI drastically reduces installer callbacks and commissioning time, protecting margins.

Technical Description

For a commercial 30kW heat pump, robust integration with building management architecture is just as important as thermodynamic efficiency. This module contains the main logic interface that communicates with the outdoor ATEX-controller via a shielded RS485 communication bus. It features a full-color capacitive touchscreen tailored for installers to conduct rapid commissioning, run automated purge routines, and view deeply granular diagnostic system telemetry.

Crucially, this module directly answers the directives of UK Building Regulations Part L, which mandate smart, energy-efficient controls. It computes 'weather compensation' formulas—reading data from an external ambient temperature sensor and mathematically lowering the water flow temperature as the outdoor weather gets warmer, drastically improving the Seasonal COP (SCOP). Furthermore, via modular Modbus or BACnet gateways, it integrates gracefully into existing SME Building Energy Management Systems (BEMS), allowing facility managers remote oversight and fleet control without traversing proprietary walled gardens.

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
Outdoor Structural & Acoustic Chassis	Corrosion in coastal UK environments	Not computed	Not computed	4×3=12	Specify marine-grade powder coating (C4/C5 classification rating)
Outdoor Structural & Acoustic Chassis	Propane pooling within chassis base	Not computed	Not computed	5×2=10	Design sloped base pan with passive bottom venting to ensure zero gas pooling
Outdoor Structural & Acoustic Chassis	Failure of MCS vibration constraints	Not computed	Not computed	4×3=12	Implement finite element analysis (FEA) to validate base rigidity prior to tooling
Air Heat Recovery Assembly (Evaporator)	Ice-expansion crushing base copper tubes	Not computed	Not computed	5×3=15	Elevate base coil and install fault-monitored 50W trace heating cable
Air Heat Recovery Assembly (Evaporator)	Inadequate thermal extraction at -5°C	Not computed	Not computed	4×2=8	Oversize evaporator coil block by 10% above baseline calculations
Air Heat Recovery Assembly (Evaporator)	Corrosion of standard aluminum fins	Not computed	Not computed	3×4=12	Specify high-build epoxy or premium hydrophilic fin coating
R290 Compressor Subsystem	Liquid refrigerant slugging on startup	Not computed	Not computed	5×3=15	Implement redundant suction accumulator and strict crankcase heating control logic
R290 Compressor Subsystem	Ignition of leaked R290 at electrical terminals	Not computed	Not computed	5×1=5	Use ATEX-certified compressor terminal housings and spark-free internal relays
R290 Compressor Subsystem	Acoustic transmission through copper lines	Not computed	Not computed	3×4=12	Incorporate discharge mufflers and double-loop vibration elimination piping

Refrigerant Routing & Control Circuit	High-pressure pipe rupture	Not computed	Not computed	5×1=5	Implement multiple mechanical pressure relief valves and 100% helium leak testing on production line
Refrigerant Routing & Control Circuit	EEV controller failure causing liquid flood-back	Not computed	Not computed	4×2=8	Implement superheat validation logic using redundant temperature sensors
Refrigerant Routing & Control Circuit	Defrost mechanism failure	Not computed	Not computed	4×3=12	Design fail-safe timing loops if temperature sensors flag non-responsive defrosts
R290-to-Water Condenser Module	Heat exchanger freezing and rupturing	Not computed	Not computed	5×2=10	Install anti-freeze thermostats and mandate 30% glycol mixture for UK warranties
R290-to-Water Condenser Module	Fouling/Scaling destroying efficiency	Not computed	Not computed	3×4=12	Require inline magnetic/dirt separators in the installation manual and warranty terms
R290-to-Water Condenser Module	Inadequate heat transfer leading to low COP	Not computed	Not computed	4×2=8	Source highly asymmetric plate heat exchangers from Tier-1 suppliers like SWEP or Alfa Laval
Outdoor ATEX-Isolated Master Controller	VFD Overheating during peak summer loads	Not computed	Not computed	4×3=12	Mount the VFD heat sink directly downstream of the primary evaporator EC cooling fans
Outdoor ATEX-Isolated Master Controller	Spark ignition of pooled R290	Not computed	Not computed	5×1=5	Utilize hermetically sealed contactors and pot all exposed high-voltage relays
Outdoor ATEX-Isolated Master Controller	Logic software bug causing compressor destruction	Not computed	Not computed	4×2=8	Implement hard-coded electromechanical safety limit switches bypassing the PCB
Indoor Hydrobox Enclosure & Manifold	Wall bracket structural failure under load	Not computed	Not computed	4×2=8	Over-engineer bracket for 200kg sheer strength and offer a floor-mount pedestal optional accessory
Indoor Hydrobox Enclosure & Manifold	Thermal heat bleed lowering SCOP	Not computed	Not computed	3×3=9	Utilize completely enclosed EPP clamshell insulation inside the steel cabinet

Indoor Hydrobox Enclosure & Manifold	Installer pipe mismatch	Not computed	Not computed	2×4=8	Standardize on universal 1-inch and 1.25-inch BSP brass threaded tails for all connections
Primary Hydronic Pump & Safety Rack	Total pump failure freezing the outdoor Condenser	Not computed	Not computed	5×2=10	Hard-wire bypass safety loops and anti-freeze valves near the outdoor unit, not just indoors
Primary Hydronic Pump & Safety Rack	System over-pressurization bursting pipes	Not computed	Not computed	4×1=4	Redundant, mechanically isolated 3-bar and 4-bar safety relief valves dumped to safe drains
Primary Hydronic Pump & Safety Rack	Inadequate flow to satisfy 30kW capability	Not computed	Not computed	4×3=12	Specify high-head commercial class Wilo/Grundfos circulating pumps capable of thick glycol viscosities
Auxiliary Electric Boiler Backup	Dry fire destruction of heating elements	Not computed	Not computed	4×2=8	Interlock heater power electrically with a fail-safe mechanical paddle flow switch
Auxiliary Electric Boiler Backup	Runaway heating boiling the water loop into steam	Not computed	Not computed	5×1=5	Install dual manual-reset bi-metallic thermal cutouts at the absolute apex of the heating tube
Auxiliary Electric Boiler Backup	Acoustic contactor "clunking" noise annoying clients indoors	Not computed	Not computed	2×4=8	Utilize totally silent zero-cross Solid State Relays (SSRs) instead of mechanical contactors
BEMS Gateway & Human Machine Interface (HMI)	Intermittent comms causing systemic shutdowns	Not computed	Not computed	4×3=12	Implement galvanic isolation on all RS485 bus lines and provide bespoke shielded comms cables in the box
BEMS Gateway & Human Machine Interface (HMI)	Cybersecurity vulnerability via networked TCP gateways	Not computed	Not computed	3×2=6	Implement highly restricted local-only networks with mandatory password rotation features

BEMS Gateway & Human Machine Interface (HMI)	Inaccurate weather compensation logic over-heating the building	Not computed	Not computed	3×3=9	Code dynamic autocalibration routines using dual interior reference thermostats
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Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	10 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success